Pavan Venkata Naga Manoj Bollepalli

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Summary

B.Tech (AI & ML) 2026 graduate with production RAG pipeline experience (Llama 3.3 70B, pgvector, O*NET), strong DSA foundations, and 1st place at ACM Programming Contest. Open-source contributor at Wikimedia Foundation. Google Cloud ACE and AWS CCP certified.

Technical Skills

Languages: Python, Java, JavaScript ,SQL, HTML/CSS
Backend: FastAPI, Node.js, Express.js, REST API Design
Frontend: Next.js, React, Tailwind CSS
Databases: PostgreSQL, pgvector (Hybrid Search), MySQL
AI / ML: Generative AI, RAG Pipelines, LLM Prompt Engineering, Vector Embeddings (Mistral)
Cloud & DevOps: AWS (Certified), Google Cloud (Certified), Docker, Linux, Git

Education

Bachelor of Technology — Computer Science & Engineering (AI & ML) Aug 2022 – May 2026
Vasireddy Venkatadri Institute of Technology (VVIT) CGPA: **8.28 / 10.0**

Experience

Open Source Developer — *Wikimedia Foundation (Remote)* Jul 2024 – Sep 2025

- Built a **community insights dashboard** in Python + Streamlit + MySQL to surface Wikipedia editor contribution trends across a global community of thousands of active contributors
- Reduced SQL query execution time by **~25%** by applying composite indexing and query restructuring on replicated production datasets
- Authored reusable Python query modules with docstrings and unit tests, eliminating repetitive ad-hoc analysis scripts used by the analytics team
- Participated in international code reviews following Wikimedia engineering standards, with contributions merged into the main repository

Projects

SkillVector — AI Career Intelligence Platform Live GitHub

- Built a **3-layer RAG retrieval pipeline** (in-memory → pgvector hybrid search → Tavily web fetch) that reduced RAG latency from **9.6s to 0.48s (95% reduction)** on warm queries; cold-start path generation completes in **~18s end-to-end** across 13 concurrent web fetches
- Implemented **hybrid vector search** using PostgreSQL + pgvector: HNSW cosine-similarity index on 1024-dim Mistral embeddings filtered by B-tree metadata (target_role), with batched UNION ALL queries reducing N DB round-trips to 1
- Integrated **O*NET 29.0** (900+ occupations) for SOC-code matching via fuzzy search with cross-domain rejection; **LLM fallback** for modern roles absent from O*NET, enabling real labor-market skill-gap analysis
- Added locking to prevent duplicate 18s generation runs when the same request fires more than once (double-clicks, retries, multiple tabs)
- **Stack:** Next.js 16, FastAPI, PostgreSQL + pgvector, Mistral Embed, Llama 3.3 70B (Groq), Tavily, O*NET 29.0

Flash Resume — Local-First AI Resume Tailoring Tool GitHub PyPI

- Cut per-application resume tailoring from **~3 min to under 10 s** by building a pip-installable tool (CLI + Chrome extension) where the LLM returns a **targeted diff** instead of a full resume, reducing generated output by **~80%** (~600 vs ~3,000 tokens)
- Prevented hallucinated experience with an **evidence gate** that checks JD keywords against the master resume and drops unsupported edits, capping rewrites at ±2 words; compiled locally with **Typst** (<100ms) and verified single-page fit via `pypdf`
- Shipped with PDF import, unit tests, and **GitHub Actions** CI/CD that publishes to PyPI on version tags
- **Stack:** Python, FastAPI, Typst, Chrome Extension (Manifest V3), GitHub Actions, Groq/Gemini APIs

Achievements & Certifications

- **AWS Certified Cloud Practitioner** — validated cloud fundamentals, security, billing, and core AWS service deployment practices
- **Google Cloud Associate Cloud Engineer** — validated expertise in cloud infrastructure, IAM, compute, networking, and scalable deployments on GCP
- **1st Place** — ACM Programming Contest (200+ participants) — awarded for algorithmic problem-solving